

Q6		
(a)(i)	Gain in kinetic energy of electron = Loss in EPE of system $\frac{1}{2}mv^2 - 0 = eV$ $v = \sqrt{\frac{2eV}{m}} = \sqrt{\frac{2(1.6 \times 10^{-19})(100)}{9.11 \times 10^{-31}}} = 5.93 \times 10^6 \text{ m s}^{-1}$	M1 A1
(a)(ii)	$\lambda = \frac{h}{p} = \frac{(6.63 \times 10^{-34})}{(9.11 \times 10^{-31})(5.93 \times 10^6)}$ $= 1.23 \times 10^{-10} \text{ m}$	M1 A1
(b)(i)	$2d \sin \theta$	B1
(b)(ii)1	The electrons emerge with a larger speed/kinetic energy and hence momentum.	B1
	By de Broglie relationship ($\lambda = \frac{h}{p}$), the wavelength of the electrons decreases.	B1
(b)(ii)	The <u>path difference</u> of the electron waves (from the different atomic planes) arriving at the detector <u>remains constant</u> , however the wavelength of the electrons decreases continually.	B1
	When the <u>path difference</u> is integer multiple of the de Broglie wavelength of the <u>electrons</u> ($0, \lambda, 2\lambda, \dots$), constructive interference occurs/ the electron waves meet in phase,	B1
	the <u>likelihood/chance/probability</u> of the electrons arriving at the detect is <u>large</u> and a maximum value of I is detected.	B1
	<i>B1 - path difference is constant</i> <i>B1 - CI /maxima occurs when path difference is integer multiple of the wavelength of the electrons.</i> <i>B1 – maxima corresponds to high chance probability of electron arriving there</i>	

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